

MindMatrix VTU Internship Program

Department of Electronics and Communication Engineering

PROJECT TITLE: 88

Android App Development using GenAI

Kreeda-Prerana Scout

(Sports Talent Tracking Application)

Android Application Project Report

Project Title	Android App Development using GenAI – Kreeda-Prerana Scout (Sports)
Project Number	88
Platform	Android (Kotlin + XML Layouts)
Domain	Sports / Physical Education / Talent Scouting
Category	Android Application with Room Database & Chronometer Engine
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1. Introduction

Kreeda-Prerana Scout is an Android application developed to help Physical Education (PE) teachers in rural and semi-urban schools digitally track and manage the athletic performance of their students. The application name combines Sanskrit and English, reflecting its purpose: to be the inspirational digital scout for every grassroots sports talent in India.

Developed using Kotlin with XML-based layouts, the app integrates Room Database for permanent on-device storage, a high-precision Chronometer engine for race timing accurate to two decimal places, and a Talent Curve analytics module to visualise student progress over time. The application targets the grassroots physical education ecosystem — a segment that has historically relied on handwritten registers and manual stopwatches to record athletic milestones such as sprint times and jump distances.

The application addresses three critical needs of PE teachers and sports scouts: creating structured digital profiles for each student athlete, logging performance data across standardised athletic tests, and generating a Talent Curve that makes it easy to identify students who have the potential to compete at district, state, or national levels. A Leaderboard feature fosters healthy internal competition, while Milestone Badges automatically recognise students who meet preset benchmarks.

Kreeda-Prerana Scout is designed to function entirely offline, requiring no internet connection for its core operations. Data is stored persistently on-device using Room Database, ensuring that performance records survive app restarts, device reboots, and long periods of inactivity. The app targets PE teachers and sports coaches in Karnataka and rural India who lack access to digital scouting platforms or talent management systems.

2. Problem Statement

India's grassroots sports talent pipeline is one of the largest and most untapped in the world, comprising millions of young athletes in rural and government schools who demonstrate exceptional physical ability but remain unnoticed due to the absence of structured performance recording. Several systemic problems affect this segment:

Challenge	Description
No Performance Records	Athletic milestones (sprint times, jump heights) are not systematically recorded, making it impossible to create a data-backed talent profile for individual students.
Talent Invisibility	Coaches and national scouts cannot identify 'Diamonds in the Rough' from rural areas without a verifiable historical performance trail.
Manual Stopwatch Errors	Handwritten timing records are prone to human error, rounding, and illegibility, undermining the credibility of performance data.
No Progress Tracking	Without longitudinal data, PE teachers cannot observe improvement trends or identify when a student crosses benchmark thresholds for district-level competition.
No Competitive Motivation	The absence of visible rankings or achievement badges limits the motivational culture needed to drive consistent athletic improvement.

These interconnected challenges highlight the urgent need for a dedicated, offline-first digital scouting application that PE teachers can use with minimal technical knowledge, on low-cost Android smartphones, without internet connectivity.

3. Project Vision

Kreeda-Prerana Scout envisions a world where every PE teacher — regardless of the school's location, budget, or technological infrastructure — has access to a reliable, intelligent, and motivational sports talent management tool. The project is built on four foundational pillars:

- **Accessibility** – Financial management tools must be available to every vendor, regardless of smartphone model, internet availability, or digital literacy. The app is designed for minimum API 21 devices and requires no internet for core features.
- **Simplicity** – The interface must be usable by PE teachers who may have limited experience with sports analytics software. Large touch targets, clear iconography, and a minimal learning curve are non-negotiable design principles.
- **Reliability** – Performance data must never be lost. Room Database provides ACID-compliant local storage that survives app crashes, device reboots, and accidental app closures.
- **Intelligence** – The burden of manual performance analysis must be reduced through technology. The Talent Curve analytics engine automatically identifies high-performing students, removing the subjectivity from talent scouting.

Kreeda-Prerana Scout is not merely a note-taking app; it is a talent empowerment tool designed to bring the discipline of structured scouting to grassroots physical education, one athletic milestone at a time.

4. Scope of the Application

4.1 Target Users

- Physical Education teachers in government and private schools across Karnataka
- Sports coaches and district-level scouts looking for structured performance data
- School administrators seeking to promote a culture of sports and physical literacy
- Khelo India programme coordinators identifying rural talent for national academies

4.2 Platform

- Android smartphones with minimum API Level 21 (Android 5.0 Lollipop and above)
- Optimised for low-end and mid-range devices with limited RAM
- APK size kept minimal to support devices with limited internal storage

4.3 Offline Functionality

- All athlete profile and performance data stored locally using Room Database — no cloud or server required
- Core features including profile creation, trial logging, Talent Curve, and leaderboard work fully offline
- Batch entry for an entire class of up to 30 students is supported in a single session

4.4 Key Operational Scenarios

- Creating a structured digital profile for a student athlete with name, age, and primary sport
- Recording a timed sprint or measured jump result using the built-in Chronometer
- Viewing the school's internal leaderboard ranked by performance scores
- Viewing the Talent Curve graph for an individual student to observe performance trends
- Awarding 'District Level Ready' milestone badges based on preset benchmark thresholds

5. Objectives

The project is aligned with the following measurable objectives:

- Provide a fast and accurate athlete profile creation system — Ensure that a new student profile can be created within one screen interaction from the home dashboard, with fields for name, age, class, gender, and primary sport.
- Deliver a high-precision trial logging system — Implement a Chronometer accurate to two decimal places (centiseconds) for timing sprints, and a distance logger for jumps and throws.
- Enable Batch Entry — Allow a PE teacher to log performance data for an entire class of 30 students within a single session without restarting the app.
- Ensure permanent, offline data persistence — All athlete profiles and trial records must survive application restarts, device reboots, and extended periods of inactivity using Room Database.
- Generate a clear and interpretable Talent Curve — Display a line graph of an individual student's performance history across multiple trials to visualise improvement trends.
- Automate Milestone Badge Awards — Automatically award 'District Level Ready' and other preset milestone badges when a student's recorded performance crosses a defined benchmark threshold.
- Provide a school-level Leaderboard — Maintain a ranked internal leaderboard by sport and event that updates instantly with every new trial entry.

6. Features

6.1 Core Features

Feature	Description
Athlete Profile	Create a digital profile for each student with name, age, class, gender, and primary sport (e.g., Kabaddi, Athletics, Long Jump). Profiles are stored permanently in Room Database.
Trial Logger	A dual-mode logger: a high-precision Chronometer (accurate to 0.01 seconds) for sprint timing, and a numeric input with unit selector (metres/centimetres) for jump and throw distances.
Batch Entry Mode	Allows a PE teacher to sequentially log performance data for an entire class roster of up to 30 students in a single session, with the student list pre-loaded from the athlete database.
Talent Curve Graph	Displays a line graph of an individual student's trial history across all recorded sessions, clearly showing performance improvement, plateaus, or regression over time.
Milestone Badges	Automatically awards achievement badges ('District Level Ready', 'State Qualifier Prospect') when a student's performance crosses preset benchmark thresholds defined by national standards.
School Leaderboard	A ranked internal school leaderboard sorted by best performance per event and sport. Provides a motivational competitive environment while keeping data within the school.
Dashboard Summary	Home screen displays total registered athletes, total trials logged, and a top-performer highlight card. Updated in real time after every trial entry.
Sport Filter	Filter athlete lists, the leaderboard, and trial history by sport or event type for focused analysis.

6.2 Future Scope Features

- **AI Performance Prediction** — Use GenAI to predict a student's performance trajectory based on historical trial data and recommend training adjustments.
- **Scout Report Export** — Generate a formatted PDF talent report for an individual student, shareable with district sports authorities.
- **Multi-Language Support** — Extend the interface to support Kannada, Hindi, Tamil, and Telugu in addition to English.
- **Cloud Sync** — Optional encrypted cloud backup via Firebase to protect data against device loss and enable multi-teacher access.
- **Video Trial Attachment** — Allow short video clips of a student's trial to be attached to the performance record for visual verification by remote scouts.
- **Biometric Login** — Replace PIN login with fingerprint authentication on supported devices.

7. Functional Requirements

The following functional requirements define the core behaviours the system must support:

FR-1: Athlete Profile Management

- The system shall allow a PE teacher to create a new athlete profile with mandatory fields: name, age, class, gender, and primary sport.
- The system shall validate that name, age, and sport fields are not empty before saving.
- The system shall store profiles in Room Database with an auto-generated unique ID and creation timestamp.
- The system shall allow viewing, editing, and deleting existing athlete profiles.

FR-2: Trial Logging

- The system shall provide a Chronometer mode for timing sprint events, accurate to two decimal places (centiseconds).
- The system shall provide a Distance Logger mode with numeric input and unit selector (metres/centimetres) for jump and throw events.
- The system shall associate each trial record with an athlete profile, event type, date, and result value.
- The system shall save the trial record to Room Database and refresh the athlete's performance history on save.

FR-3: Batch Entry

- The system shall provide a Batch Entry screen pre-loaded with the class roster of registered athletes.
- The system shall allow the teacher to sequentially enter a performance result for each student without returning to the main screen.
- The system shall save all batch entries on a single confirmation action at the end of the session.

FR-4: Talent Curve

- The system shall display a line graph of an individual athlete's trial results across all recorded sessions for a selected event.
- The graph shall clearly label axes with trial dates (X-axis) and performance values (Y-axis).
- The system shall highlight benchmark threshold lines on the graph for comparison against milestone standards.

FR-5: Milestone Badges

- The system shall define preset benchmark values for each event type (e.g., 100m sprint under 13.5 seconds = 'District Level Ready').
- The system shall automatically evaluate each new trial result against applicable benchmarks.
- The system shall award and display a milestone badge on the athlete's profile when a benchmark is met or exceeded.

FR-6: Leaderboard

- The system shall maintain a ranked leaderboard of athletes sorted by best performance per event.
- The leaderboard shall update instantly after every new trial entry.
- The system shall allow filtering the leaderboard by sport and event type.

8. Non-Functional Requirements

Category	Requirement	Acceptance Criterion
Performance	Dashboard Load Time	Dashboard must fully render within 800 ms of app launch on a mid-range Android device.
Performance	Trial Save Speed	A trial result must be saved and reflected in the athlete's history within 500 ms of tapping Save.
Timer Accuracy	Chronometer Precision	The Chronometer must display and record elapsed time to exactly two decimal places (centiseconds).
Usability	UI Simplicity	Any PE teacher must be able to log a trial for an existing athlete within 3 taps from the home dashboard.
Usability	Legibility	All primary text must use a minimum font size of 13sp. Leaderboard values must use a minimum of 20sp.
Reliability	Data Persistence	100% of athlete profiles and trial records must be retained after app closure, device restart, and 30+ days of inactivity.
Batch Entry	Class Size Support	Batch Entry must support logging performance for a full class of 30 students without error or crash.
Compatibility	Android Versions	Application must support Android 5.0 (API 21) through Android 14 (API 34).
Offline Support	Core Features	All core features (profiling, trial logging, Talent Curve, leaderboard) must function without any network connection.

9. System Architecture

Kreeda-Prerana Scout follows a single-tier, client-only architecture. All data processing, storage, business logic, and analytics occur entirely on-device. There are no backend servers, cloud databases, or API dependencies required for the application's core operation.

9.1 Technology Stack

Layer	Technology	Purpose
Language	Kotlin	Primary development language; concise, null-safe, fully interoperable with Android SDK.
UI Framework	XML Layouts + AppCompat	Traditional View-based UI with LinearLayout, RecyclerView, and Material components for broad device compatibility.
Local Database	Room Database (SQLite)	ACID-compliant ORM over SQLite. Provides permanent on-device storage with DAO-based queries and migration support.
Charting	MPAndroidChart / Canvas API	Renders the Talent Curve line graph with benchmark threshold annotations and interactive data points.
Timer Engine	Android Chronometer + Handler	High-precision lap timer leveraging SystemClock.elapsedRealtime() for centisecond-accurate sprint timing.
Session Storage	SharedPreferences	Stores benchmark threshold configurations, app preferences, and filter settings as lightweight key-value pairs.
Build System	Gradle (Groovy DSL)	Dependency management, build configuration, and APK generation including kotlin-kapt for Room annotation processing.
Min. SDK	API Level 21	Targets approximately 98% of active Android devices globally, including low-cost smartphones prevalent in rural India.

9.2 Architectural Overview

The application follows a simplified MVC (Model-View-Controller) pattern appropriate for a single-activity, XML-layout Android app:

- **Model Layer** – The data package contains the Room @Entity classes (Athlete, TrialRecord), @Dao interfaces (AthleteDao, TrialDao), and @Database singleton (AppDatabase). This layer owns all data access logic and migration handling.
- **View Layer** – XML layout files define the UI structure (activity_main.xml, activity_athlete_profile.xml, activity_trial_logger.xml, activity_talent_curve.xml, activity_leaderboard.xml, list_item_athlete.xml, list_item_trial.xml). Custom drawables provide visual styling including sport-category icons and milestone badge graphics.

- **Controller Layer** – Activity and Fragment classes (MainActivity, AthleteProfileActivity, TrialLoggerActivity, TalentCurveActivity, LeaderboardActivity, BatchEntryActivity) orchestrate user interaction, invoke database operations on background threads via Executors, and update the UI on the main thread via `runOnUiThread`.

9.3 Database Design

The Room database contains two primary tables — athletes and trial_records — with the following schema:

Table: athletes

Column	Type	Constraint	Description
id	INTEGER	PRIMARY KEY, AUTO	Auto-generated unique athlete identifier.
name	TEXT	NOT NULL	Full name of the student athlete.
age	INTEGER	NOT NULL	Age of the athlete at the time of registration.
gender	TEXT	NOT NULL	Gender of the athlete (Male/Female/Other).
sport	TEXT	NOT NULL	Primary sport category (e.g., Athletics, Kabaddi).
badges	TEXT	DEFAULT ""	JSON array of awarded milestone badge identifiers.
created_at	TEXT	NOT NULL	Formatted date and time of profile creation.

Table: trial_records

Column	Type	Constraint	Description
id	INTEGER	PRIMARY KEY, AUTO	Auto-generated unique trial identifier.
athlete_id	INTEGER	NOT NULL, FK	Foreign key referencing athletes.id.
event_type	TEXT	NOT NULL	Event name (e.g., 100m Sprint, Long Jump).
result	REAL	NOT NULL	Numeric performance value (seconds or metres).
unit	TEXT	NOT NULL	Unit of measurement: 'seconds', 'metres', or 'cm'.
recorded_at	TEXT	NOT NULL	Formatted date and time of trial recording.

10. User Flow

The application is designed to minimise friction in high-frequency daily-use scenarios such as conducting a PE class trial session or updating the school leaderboard after an inter-house competition. The complete user journey is described below:

#	Screen	User Action & System Response
1	Home Dashboard	Teacher launches app. Home screen displays total registered athletes, total trials logged, and a top-performer highlight card.
2	Add Athlete	Teacher taps '+' button. Fills in student name, age, class, gender, and primary sport. Taps Save. Profile stored in Room DB and athlete appears in the list.
3	Trial Logger	Teacher selects an athlete and taps 'Log Trial'. Selects event type (sprint or distance). For sprint: taps Start/Stop on the Chronometer. Result auto-populated. Taps Save Trial.
4	Batch Entry	Teacher taps 'Batch Entry'. Class roster pre-loaded. Teacher enters result for each student sequentially. Confirms at end. All records saved to Room DB in a single transaction.
5	Talent Curve	Teacher opens an athlete's profile and taps 'View Talent Curve'. A line graph displays all recorded trial results with benchmark threshold lines overlaid.
6	Milestone Badge	On saving a trial result that crosses a benchmark (e.g., 100m in < 13.5 s), the system automatically awards a 'District Level Ready' badge and displays it on the athlete's profile.
7	Leaderboard	Teacher taps 'Leaderboard'. A ranked list of all athletes sorted by best performance for a selected event is displayed. Teacher can filter by sport category.
8	Delete / Edit	Teacher long-presses an athlete profile or trial record. A confirmation dialog appears. On confirmation, the record is deleted and lists and summaries update instantly.

11. Technical Implementation

11.1 Room Database & Migration

Athlete profiles and trial records are stored using the Room persistence library, which provides a type-safe abstraction over SQLite. The AthleteDao interface exposes operations: `getAll()`, `insert()`, `update()`, `delete()`, and `getById()`. The TrialDao exposes `getByAthleteId()` (ordered by id DESC to show newest first), `insert()`, `delete()`, and `deleteByAthleteId()` for cascading deletes. The AppDatabase singleton uses a double-checked locking pattern to ensure a single database instance throughout the app lifecycle.

11.2 High-Precision Chronometer Engine

The sprint timer uses `SystemClock.elapsedRealtime()` to capture start and stop timestamps with millisecond precision. A Handler post-delayed loop refreshes the display every 10 ms, rendering elapsed time in MM:SS.cc format where cc represents centiseconds. The recorded result is stored as a REAL (Double) value in the `trial_records` table in seconds with two decimal places, satisfying the success criterion of timer accuracy.

11.3 Talent Curve Analytics

The Talent Curve screen queries the TrialDao for all trial records associated with a given `athlete_id` and `event_type`, ordered by `recorded_at` ascending to produce a chronological performance series. The data is rendered as a line graph using MPAndroidChart's LineChart widget. Preset benchmark threshold values are drawn as horizontal LimitLine annotations, visually separating 'Below Standard', 'District Level Ready', and 'State Qualifier Prospect' zones.

11.4 Batch Entry System

The BatchEntryActivity loads the full athlete roster into a RecyclerView where each row contains the athlete's name and an input field for the trial result. Input validation ensures non-empty and positive numeric values for all entries before the 'Save All' action is enabled. On confirmation, all trial records are inserted into Room DB within a single background Executor call wrapped in a database transaction to ensure atomicity — either all 30 records are saved or none.

11.5 Milestone Badge Engine

Benchmark thresholds are defined as constants in a BenchmarkConfig object keyed by event type and gender. After every trial save, the system automatically compares the recorded result against all applicable benchmarks. If a threshold is crossed for the first time, the corresponding badge identifier is appended to the athlete's badges JSON array field in the athletes table and the profile UI refreshes to display the new badge.

11.6 Background Threading

Room does not permit database operations on the main (UI) thread. All database calls are executed using a single-threaded `Executors.newSingleThreadExecutor()` instance. Results are posted back to the UI thread via `runOnUiThread()`, which updates RecyclerView adapters and summary TextViews. This pattern avoids the need for LiveData or coroutines while maintaining thread safety.

12. Project File Structure

File / Directory	Purpose
app/build.gradle	App-level Gradle config: plugins, SDK versions, Room, MPAndroidChart, and kapt dependencies.
data/Athlete.kt	Room @Entity data class representing a student athlete profile.
data/TrialRecord.kt	Room @Entity data class representing a single athletic trial result.
data/AthleteDao.kt	Room @Dao interface with getAll, getById, insert, update, and delete queries.
data/TrialDao.kt	Room @Dao interface with getByAthleteId, insert, delete, and deleteByAthleteId queries.
data/AppDatabase.kt	Room @Database singleton with version management and migration definitions.
config/BenchmarkConfig.kt	Defines benchmark threshold constants for each event type and gender category.
MainActivity.kt	Home dashboard: summary cards, navigation to all major screens, athlete list.
AthleteProfileActivity.kt	Athlete profile form: name, age, class, gender, sport. Displays awarded badges.
TrialLoggerActivity.kt	Trial logging screen with Chronometer mode (sprint) and Distance Logger mode (jumps/throws).
BatchEntryActivity.kt	Batch entry screen with class roster RecyclerView and bulk save functionality.
TalentCurveActivity.kt	Talent Curve screen rendering MPAndroidChart line graph with benchmark LimitLines.
LeaderboardActivity.kt	School leaderboard screen with ranked athlete list, sport filter, and best-result display.
AthleteAdapter.kt	RecyclerView adapter inflating list_item_athlete.xml for athlete roster display.
TrialAdapter.kt	RecyclerView adapter inflating list_item_trial.xml for trial history display.
res/layout/activity_main.xml	Dashboard layout: header, summary cards, athlete list, FAB for new athlete.
res/layout/activity_trial_logger.xml	Trial logger layout: Chronometer display, Start/Stop button, event selector, distance input.
res/layout/activity_talent_curve.xml	Talent Curve layout: MPAndroidChart LineChart widget, event filter spinner.

res/layout/activity_leaderboard.xml	Leaderboard layout: ranked RecyclerView, sport/event filter chips.
res/drawable/*.xml	Shape drawables for card backgrounds, avatar circles, badge icons, and sport category icons.
AndroidManifest.xml	App manifest: permissions, activity declarations, and launcher activity definition.

13. Impact Goals

13.1 Khelo India Talent Pipeline

By creating structured, data-backed athlete profiles for students in rural government schools, Kreeda-Prerana Scout directly supports the Khelo India programme's mission of identifying grassroots talent early. Scouts and national academy selectors can request digitally recorded performance data instead of relying on unverified verbal assessments, increasing the chances of rural talent receiving national-level opportunities.

13.2 Physical Literacy

The daily use of Kreeda-Prerana Scout by PE teachers creates a culture of structured, record-based physical education. Students who see their performance recorded and compared against benchmarks develop a growth mindset toward physical activity, encouraging consistent participation and improvement.

13.3 Equal Opportunity

A child in a remote village now has the same quality of performance documentation as a student in an elite urban sports academy. The digital record levels the playing field — talent is identified by data, not by geography or connections.

13.4 Motivational Culture

The internal school leaderboard and milestone badge system create a healthy, transparent competitive environment. Students are motivated to improve their personal bests, aware that their performances are tracked and visible. This competitive spirit, channelled positively, drives athletic development at the grassroots level.

13.5 Scalability of Impact

The modular architecture of Kreeda-Prerana Scout supports rapid expansion. Future versions can add GenAI performance prediction, PDF scout report export, cloud sync, and video trial attachments without requiring a fundamental redesign of the existing codebase.

14. Success Criteria

The project will be considered successfully completed when the following criteria are met:

#	Success Criterion	Measurement	Priority
1	App loads and reaches the Dashboard within 800 ms on a reference device (Redmi 9A or equivalent).	Android Profiler	Critical
2	A trial for an existing athlete can be logged within 3 taps from the home dashboard.	Manual UI walkthrough	Critical
3	Chronometer displays and records sprint time accurate to two decimal places (centiseconds).	Reference stopwatch comparison	Critical
4	Batch Entry successfully logs performance for a full class of 30 students without error or crash.	Manual batch session test	Critical
5	All athlete profiles and trial records are retained after device reboot and 30+ days of inactivity.	Room DB inspection test	Critical
6	Talent Curve graph is clear, correctly labelled, and easy to interpret for non-technical users.	PE teacher user testing	High
7	Milestone badge awarded automatically and correctly when a benchmark threshold is crossed.	Manual benchmark test	High
8	Leaderboard updates correctly after every trial entry and deletion, with correct ranking.	Manual validation	High
9	App does not crash during a complete walkthrough of all screens and features.	Manual regression testing	High
10	All core features work fully in Airplane Mode (no internet connection required).	Airplane Mode test	Medium

Conclusion

Kreeda-Prerana Scout represents a focused and socially meaningful application of Android development in the domain of grassroots sports talent scouting and physical education management. By addressing a clear and well-defined problem — the absence of a structured, offline, and intelligent performance tracking tool for PE teachers in rural India — the project delivers measurable practical value alongside strong technical merit.

The application's architecture prioritises reliability, precision, and offline-first operation above all else. Every design decision — from Room Database for permanent storage to the SystemClock-based Chronometer for centisecond accuracy, from the MPAndroidChart Talent Curve to the BenchmarkConfig-driven badge engine — is motivated by a single question: will this work for a PE teacher standing on a dusty school ground with a basic Android smartphone and no internet connection?

The project also establishes a strong technical foundation for future expansion. The separation of data, view, and controller layers supports the addition of new features — GenAI performance prediction, PDF scout report export, cloud sync, multi-language support — without requiring a fundamental redesign of the existing codebase. The Room migration framework ensures that future schema changes can be deployed to existing users without data loss.

In conclusion, Kreeda-Prerana Scout is not simply a technical deliverable — it is a small but meaningful step toward equal opportunity and digital empowerment for the millions of young athletes in rural India who deserve the chance to be discovered, recognised, and given the platform to compete at their highest potential.